

S3 Supplementary QC and Robustness

Support, artifact, physiology, and modality-specific audits

Short title: S3 QC/Robustness - NeuralManifoldDynamics

Robin Langell¹

¹ Langell Konsult AB, Vallentuna, Sweden

Corresponding author: Robin Langell, hello@noeticdiffusion.com

Keywords: QC, robustness, geometry, HRV, MEG

S3.0 Sample and epoch coverage overview

Coverage is reported separately for completed exports and inferential subsets. The numbers below are not interchangeable:

Run or cohort	Completed units	Coverage evidence	Inferential sample
ds004511 reference	132 HDF5 outputs	45 subjects, 3 tasks	not a contrast package
ds003838 full export	130 HDF5 outputs	6,543 blocks; 27,670 windows	N = 62 complete-case stages
ds003645 MEG pilot	5 subjects	8/4/2 s window checks	pilot only
ds006848 verbal WM	42,593 feature rows	30 subjects; event-locked QC	n = 30 for listed contrasts
DANDI 000006 Units	1 session	1,752 mnps_3d windows	transport/descriptive only
DANDI 000458 LFP	2 probes, 1 session	3,433 epochs/probe; 315/384 and 200/384 channel pass	transport/descriptive only

The ds003838 full export and corrected statistics package use different run directories. The latter reports 26,460 windows and N = 62; the former reports 27,670 windows and 130 completed HDF5 subjects.

S3.1 Physiological QC pass rates

The ds006848 validation found inverted ECG polarity in 39,477 of 42,593 rows (92.7%). After polarity correction, the population median HR was 76.2 bpm and RMSSD was 40.7 ms. The same validation found retrieval-event overlap in 37,352 of 42,593 HRV windows (87.7%). Only approximately 2% of working-memory windows passed the strict clean HRV gate; rest-phase windows had an approximately 85% pass rate. Working-memory HRV inference was consequently withheld.

The ds003838 full run recorded HRV artifact, coverage, and quality columns in the block-native sidecars and preserved `anchor_quality`. These are contract and QC outputs; they do not by themselves prove physiology-coordinate causality.

S3.2 EEG/EOG artifact balance across tasks

The ds006848 artifact follow-up found a condition-associated frontal 30–45 Hz contrast ($\chi^2 = 23.00$, $p = 0.00004$) whose direction was opposite to the manifold m/d ordering. Temporal muscle power was not significant ($p = 0.61$). This narrows, but does not eliminate, high-frequency artifact as an alternative explanation. ICA/RANSAC cleaning was not run for that claim package, so the result must not be described as ICA-clean EEG.

Blink and EOG quality are retained as feature-level covariates and QC fields. A single consolidated blink-rate table is not archived in the current article evidence pack; no missing values are imputed here.

S3.3 Robustness and confound audit

Intersection sample and epoch matching

For ds003838, the full block-native join matched source windows with fraction 1.0. Statistical contrasts were restricted to $N = 62$ subjects with complete stage coverage. For ds006848, event-window purity was evaluated with the 04c filters; short events and overlapping 8 s windows make some phases unsuitable for direct interpretation.

Leave-one-subject-out and bootstrap

The ds006848 encoding m and d contrasts remained significant in the cohort-specific leave-one-subject-out analysis for 30/30 omissions. Bootstrap intervals and directional consistency were reported in the 04e follow-up, with approximately 67–77% directional consistency depending on the metric. These checks belong to the ds006848 package and are not a blanket robustness claim for every modality.

Within-subject permutation

The ds003645 pilot used label-shuffle nulls for subject-level event-response cosines. Three of five subjects had observed values above their null means, and two had shuffle $p < 0.05$; the d-family sign reversal in other subjects created a bimodal cross-modal pattern. The later holdout found no matched EEG–MEG residual topographic correspondence beyond its nulls.

Artifact balance

The ds006848 frontal high-frequency result pointed opposite to the manifold effect and temporal muscle power was non-significant, but missing ICA remains an explicit limitation. MEG

analyses require features_projection_z and per-row row_source; historical positional half-split behavior is not an acceptable provenance substitute.

Physiological confound controls

The ECG polarity correction and HRV contamination gates are measurement validity controls. They do not transform HRV into a phase-pure task measure. PPG and pupil quality are dataset-dependent. For DANDI 000458, channel pass counts, stimulation midpoint guards, contact count, and reference choice are reported as one-session sensitivity information; no cross-probe geometry claim is made.

S3.4 Follow-ups

The repository contains several named follow-up tracks rather than the template's generic Script A–D sequence.

S3.4a Event purity and physiological follow-up

Scripts 04c_window_overlap_purity_audit.py, pr01_eeg_artifact_audit.py, pr03_shortwindow_robustness.py, and pr04_artifact_balance.py evaluate window overlap, artifact proxies, shorter-window feasibility, and artifact balance for ds006848. Their outputs are cohort-specific and preserve the decision to withhold WM-phase HRV.

S3.4b MEG readiness and holdout

The ds003645 pilot passed the HDF5 contract, feature completeness, window robustness, and Jacobian-validity gates, yielding readiness 0.7879. This is below the 0.80 expansion threshold and was labelled usable pilot mapping. The holdout's residual topographic correspondence was negative, so the cross-modal claim remains exploratory.

S3.4c Conventional-feature baseline

The classical EEG comparator pack reports bandpower, slowing ratios, complexity, and connectivity summaries beside MNPS features. In ds006848, the frontal theta ordering was approximately inverse to the MNPS m/d ordering, a useful dissociation. This is a descriptive baseline comparison, not a test of incremental predictive validity.

S3.4d DANDI, LFP, and infant EEG transport audits

DANDI 000006 Units and 000458 LFP paths complete smoke or one-session QC checks. Figshare BDF ingest completed a 71-file run with 5,273 retained epochs and 70 coverage-passing outputs, but behavioural condition labels were not recoverable from the available Status field. These results document transport and QC, not clinical or circuit-level manifold inference.

Units (000006, 000009). The 000006 smoke session (sub-anm369962) produced 1,752 windowed mnps_3d rows and a full (1750, 3, 3) Jacobian export with zero pipeline errors, using the dedicated ephys-to-9D subcoordinate map described in S1.3. This is read as a starter mapping, not validated manifold evidence: the 000009 inventory run (10 units, 32 electrodes) found unit-quality labels unresolved (unknown) rather than pass/fail.

LFP (000458). A channel-count smoke run processed 32 channels into 28 epochs. The full-session two-probe QC pass retained 315/384 channels for probe B (48 depth-balanced contacts selected) and 200/384 channels for probe F (32 contacts selected), yielding 3,433 epochs per probe. State/stimulation-aware epoching split each probe's epochs across awake, isoflurane, and recovery blocks (605/438/794 epochs respectively, with a 1 s stimulation guard). A contact-count/reference-choice sensitivity check compared an 8-contact native-reference configuration against 4-contact

and median-within-probe-reference variants; the 4-contact/median-reference cell was not run, and cross-run mnps_3d coordinate orientation was not identified between configurations.

Readiness gate. Unlike the MEG pilot (S3.4b), no weighted readiness score has been computed for either the Units or the LFP ephys path. Both are scoped as transport-and-descriptive sensitivity checks. Both 000006 and 000458 are rodent recordings; no cross-species claim is made relative to the human EEG/MEG/fMRI reference paths.

S3.5 Reviewer-facing robustness summary

Robustness summary

The strongest supported statements are contract-level: valid support is recorded, source-window joins can be audited, and modality-specific failure modes are surfaced. The ds003838 HRV/block-native export and its complete-case stage contrasts survive the stated corrected-ECG pipeline, while ds006848 working-memory HRV is explicitly gated out by contamination. The MEG path remains an exploratory pilot below its readiness threshold, and the later residual-topography holdout is negative. Chart invariance, independent replication, clinical prediction, and a universal cross-modality robustness claim remain unestablished.

S3.6 Synthetic estimator and coordinate robustness

The analysis-owned CHART and M3 extension studies exercise support and provenance contracts separately from empirical cohort QC. Evaluation targets are excluded from cross-fit labels; fixed held-out queries, neighbour indices, and observed raw-increment covariance are shared across comparators. Padding and inter-segment increments are excluded, and target-only

perturbations leave both candidate predictions unchanged. Undefined statistics retain a validity reason rather than being replaced by zero.

The fixed mixed orthogonal rotation and isotropic scaling preserve the expected vector transformations, with maximum discrepancy approximately 2.9×10^{-14} . Anisotropic scaling changes Euclidean neighbours and estimator outputs (maximum discrepancies approximately 2.88 for local mean and 4.96 for local linear); the oracle coordinate identity remains valid. The associated alignment and effective-dimension changes are metric dependence expected under anisotropic scaling, not failed coordinate invariance. The unregularized CHART alignment diagnostic is distinct from the epsilon-regularized production C1 quantity. A 3D-to-2D toy projection has identical projected current states but different projected expected futures because of a hidden coordinate; this is an information-loss counterexample, not a failed coordinate transformation.

The M3 extension's six cases all completed with the fixed-query design reported in S2.8. Its order-corruption controls are labelled as artificial cross-segment controls, not temporal nulls. The finite synthetic predictions therefore support implementation and contract checks only. They do not qualify empirical drift, an Itô drift, Markov sufficiency, a production C1 object, a biological interpretation, or a universal estimator choice. The separate W_Q sensitivity diagnostic on the same sub-99 night computed 864/864 planned comparisons in 3D and 845/864 in 9D, with 19 `post_dtype_psd_failure` cases; 9D `v_norm` remains METHOD-LIMITED. These overlapping same-night perturbation trials are not 864 independent observations, do not select a production horizon, and do not close a biological gate. The source full-night 9D `status_reachability_numerical_overflow` remains retained separately; the synthetic CHART/M3 checks do not override it.

Bibliography

Competing Interests

The authors have declared that no competing interests exist.